DIPARTIMENTO DI ELETTRONICA, INFORMAZIONE E BIOINGEGNERIA

Politecnico di Milano

Machine Learning (Code: 097683)

February 1, 2023

Surname:
Name:
Student ID:
Row: Column:

Time: 2 hours Maximum Marks: 33

- The following exam is composed of **8 exercises** (one per page). The first page needs to be filled with your **name, surname and student ID**. The following pages should be used **only in the large squares** present on each page. Any solution provided either outside these spaces or **without a motivation** will not be considered for the final mark.
- During this exam you are **not allowed to use electronic devices**, such as laptops, smartphones, tablets and/or similar. As well, you are not allowed to bring with you any kind of note, book, written scheme, and/or similar. You are also not allowed to communicate with other students during the exam.
- The first reported violation of the above-mentioned rules will be annotated on the exam and will be considered for the final mark decision. The second reported violation of the above-mentioned rules will imply the immediate expulsion of the student from the exam room and the **annulment of the exam**.
- You are allowed to write the exam either with a pen (black or blue) or a pencil. It is your responsibility to provide a readable solution. We will not be held accountable for accidental partial or total cancellation of the exam.
- The exam can be written either in **English** or **Italian**.
- You are allowed to withdraw from the exam at any time without any penalty. You are allowed to leave the room not early than half the time of the duration of the exam. You are not allowed to keep the text of the exam with you while leaving the room.
- **Three of the points will be given on the basis on how quick you are in solving the exam.** If you finish earlier than 30 min before the end of the exam you will get 3 points, if you finish earlier than 20 min you will get 2 points and if you finish earlier than 10 min you will get 1 point (the points cannot be accumulated).
- The box on Page 10 can only be used to complete the Exercises 7, and/or 8.

Ex. 1	Ex. 2	Ex. 3	Ex. 4	Ex. 5	Ex. 6	Ex. 7	Ex. 8	Time	Tot.
/ 7	/ 7	/ 2	/ 2	/ 2	/ 2	/ 4	/ 4	/ 3	/ 33

Exercise 1 (7 marks)

Illustrate the AdaBoost algorithm, explain its purpose and in which case it is useful.

Exercise 2 (7 marks)

Describe and compare Ridge regression and Lasso regression.

Exercise 3 (2 marks)

Consider the following snippet of code:

```
1 X = zscore(dataset[['sepal-length', 'sepal-width']].values)
2 t = dataset['class'].values == 'Iris-setosa'
3 X, t = shuffle(X, t, random_state=0)
4
5 w = np.ones(3)
6 for i, (x_i, t_i) in enumerate(zip(X, t)):
7   ext_x = np.concatenate([np.ones(1), x_i.flatten()])
8   if np.sign(w.dot(ext_x)) != t_i:
9   w = w + ext_x * t_i
```

1. Describe the procedure presented above. What is the purpose of such a procedure? Which problem it solves?

2. Tell if the procedure above is correct and, in the case it is not, propose a modification to fix it.

3. Do you think that Line 3 is fundamental for this procedure? Can you describe an example where it can be removed and motivate why it is not useful in such a case?

1. The snippet is applying the perceptron online learning algorithm to the iris dataset. At first, it standardize the data and shuffle the samples. Then it initialize the weight vector $w = [111]$ and apply the gradient descent update over each of the samples in the training set. It solves a binary classification problem.

2. The procedure should be repeated multiple times over the dataset. In this case, a single iteration over the dataset does not guarantee that the algorithm converges to a meaningful solution, even if the dataset is linearly separable. We should add a for loop to perform the procedure multiple times over the dataset, until the solution w does not change for an entire epoch (an iteration over the entire dataset) or a maximum number of iterations is reached. In addition, the encoding of the target variable (line 2) is not correct. The target variable should be encoded as -1 and 1 instead of 0 and 1.

3. Here it is important to shuffle the data since it might occur that the samples are ordered (e.g., the ones of the positive class are all in the first positions of the dataset), which might slow down the optimization procedure. The same holds if we want to apply a train/validation approach. Instead, if we are using a closed form solution, like the one for linear regression, the shuffling procedure is not mandatory.

Exercise 4 (2 marks)

Tell if the following statements about SVM are true or false. Motivate your answers.

1. There exists a closed form solution to provide the optimal weights of the SVM.
2. There exists a unique solution for the problem of optimizing the weights of the SVM.
3. Since they have the same activation function, it is equivalent to train an SVM and a perceptron, if we have a linearly separable dataset.
4. The boundary of a nonlinear kernel SVM is linear in a specific high-dimensional feature space.

1. FALSE The problem of finding the optimal parameter requires to transform the primal optimization problem into the dual one and apply an iterative procedure (i.e., SMO) to find the optimal weights.
2. TRUE The objective function of the SVM is convex, therefore it has a unique minimum.
3. FALSE Even if they have the $sign(\cdot)$ activation function, they are optimizing different objectives. The perceptron is minimizing the misclassification error, while the SVM is maximizing the intraclass margin. This holds also for linearly separable datasets.
4. TRUE The idea of using a kernel is based on the idea that expanding the features into a complex space it is more likely the data can be all correctly classified by a straight boundary (i.e., a hyperplane).

Exercise 5 (2 marks)

You are given with the following class of regression models $\mathcal{M}_\psi$ over the variables $\{x_1, x_2\}$ and an additional feature $\psi(x_1, x_2)$:

$$\mathcal{M}_\psi : \qquad y_{\mathbf{w}}(x_1, x_2) = w_0 + w_1 x_1 + w_2 x_2^2 + w_3 \psi(x_1, x_2), \qquad \mathbf{w} \in \mathbb{R}^4.$$

Consider three possible choices of feature ψ:

$$\psi_1(x_1, x_2) = 1, \qquad \psi_2(x_1, x_2) = x_1 - 3x_2^2, \qquad \psi_3(x_1, x_2) = x_1 x_2.$$

Tell if the following statements are true or false. Motivate your answers.

1. Models $\mathcal{M}_{\psi_1}$ and $\mathcal{M}_{\psi_2}$ have the same bias.
2. The bias of model $\mathcal{M}_{\psi_1}$ is larger than the bias of model $\mathcal{M}_{\psi_3}$.
3. The VC dimension of model $\mathcal{M}_{\psi_2}$ is larger than the VC dimension of model $\mathcal{M}_{\psi_3}$.
4. The VC dimension of model $\mathcal{M}_{\psi_1+\psi_2}$ is larger than the VC dimension of model $\mathcal{M}_{\psi_3}$.

1. TRUE. Indeed, they are the very same model after a renaming of the weights:

$$\begin{aligned}\mathcal{M}_{\psi_1} : \quad y_{\mathbf{w}}(x_1, x_2) &= w_0 + w_1 x_1 + w_2 x_2^2 + w_3 \\ &= \underbrace{w_0 + w_3}_{w_0^{(1)}} + \underbrace{w_1}_{w_1^{(1)}} x_1 + \underbrace{w_2}_{w_2^{(1)}} x_2^2, \qquad \mathbf{w}^{(1)} \in \mathbb{R}^3,\end{aligned}$$

$$\begin{aligned}\mathcal{M}_{\psi_2} : \quad y_{\mathbf{w}}(x_1, x_2) &= w_0 + w_1 x_1 + w_2 x_2^2 + w_3 (x_1 - 3x_2^2) \\ &= \underbrace{w_0}_{w_0^{(2)}} + \underbrace{(w_1 + w_3)}_{w_1^{(2)}} x_1 + \underbrace{(w_2 - 3w_3)}_{w_2^{(2)}} x_2^2, \qquad \mathbf{w}^{(2)} \in \mathbb{R}^3.\end{aligned}$$

2. TRUE. Indeed, models $\mathcal{M}_{\psi_3}$ are a super-set of models $\mathcal{M}_{\psi_1}$. Formally:

$$\begin{aligned}\mathcal{M}_{\psi_1} : \quad y_{\mathbf{w}}(x_1, x_2) &= w_0 + w_1 x_1 + w_2 x_2^2 + w_3, \\ \mathcal{M}_{\psi_3} : \quad y_{\mathbf{w}}(x_1, x_2) &= w_0' + w_1' x_1 + w_2' x_2^2 + w_3' x_1 x_2.\end{aligned}$$

   To get $\mathcal{M}_{\psi_1}$ from $\mathcal{M}_{\psi_3}$, simply set $w_0' = w_0 + w_3$, $w_1' = w_1$, $w_2' = w_2$, and $w_3' = 0$.

3. FALSE. For the same reason as the previous question, recalling that models $\mathcal{M}_{\psi_2}$ are the same as models $\mathcal{M}_{\psi_1}$.
4. FALSE. Indeed, models $\mathcal{M}_{\psi_1+\psi_2}$ are the same as models $\mathcal{M}_{\psi_1}$ (or models $\mathcal{M}_{\psi_2}$).

Exercise 6 (2 marks)

We are given the problem of evaluating the time and materials necessary for the construction of a building. Even if in principle all the quantities are known, the fact that there might be some delays in the construction site and the fact that some of the materials might be lost during transportation, allow us to consider all the quantities that are involved in the process as random variables. Let us assume to have a dataset of previous construction building, where they describe the entire process, also including the intermediate steps, e.g., design, foundation building, structure building, furnishing of the apartments.

1. Model the problem of evaluating the time and materials used for a new construction.
2. Model the problem of selecting the best way of performing the construction works to minimize the time and materials required for the construction.

In both cases, specify the problem, feature, solution and all the relevant information to solve such task using ML-based approaches.

1. Multiple regression problem: we can use as input the characteristics of the building ground, the one about the location we are planning to built to predict the amount of materials and time required (output) to finish the building. A method to solve such a problem is to use multiple linear regression. Notice that, with this approach, we are only considering the process as a whole, while we are not taking into account the intermediate steps of the process.
2. Reinforcement Learning problem: given the actions required to finish a building, select the ones which are the most fruitful from a time and economic point of view. The state consists in the current state of the building, the actions are the decision one may take to advance the building construction, the reward is either time or costs incurred by the specific action, the transition are stochastic, the discount factor might be set to 1 since we are assuming to have a finite horizon for the process. We can use either SARSA or Q-learning to find the optimal policy. If we want to make use of the dataset we need to use an offline offpolicy approach or an offline approach with importance sampling.

Exercise 7 (4 marks)

Consider the set of trajectories collected with a policy π in a Markov decision process with state space $\mathcal{S} = \{A, B, C\}$, where C is an absorbing zero-reward state, and action space $\mathcal{A} = \{x, y\}$:

$$(A, x, -2) \to (B, y, 3) \to (A, x, 1) \to (C)$$
$$(B, x, 1) \to (B, y, -3) \to (C)$$
$$(A, y, 6) \to (C)$$

1. Execute first-visit Monte Carlo for estimating the value function $V^\pi(s)$ for all states $s \in \mathcal{S}$, keeping the discount factor γ parametric.
2. Execute TD(0) **on the first trajectory only** for estimating the value function V^π for all states $s \in \mathcal{S}$, keeping the discount factor γ parametric, starting with a zero initialization of the value functions and with learning rate $\alpha = 1$.
3. Can the provided trajectories be generated by a deterministic policy π? Justify your answer.

1. Let us apply first-visit Monte Carlo:

$$V^\pi(C) = 0 \text{ (absorbing zero-reward state)}$$
$$V^\pi(B) = \frac{1}{2}\left((3+\gamma) + (1-3\gamma)\right) = 2 - \gamma$$
$$V^\pi(A) = \frac{1}{2}\left((-2+3\gamma+\gamma^2) + 6\right) = 2 + \frac{3}{2}\gamma + \frac{1}{2}\gamma^2.$$

2. Let us apply TD(0) starting with a zero initialization of the value functions and with learning rate $\alpha = 1$:

$$(A, x, -2) \to (B, y, 3) \implies V^\pi(A) \leftarrow R + \gamma V^\pi(B) = -2 + \gamma \cdot 0 = -2$$
$$(B, y, 3) \to (A, x, 1) \implies V^\pi(B) \leftarrow R + \gamma V^\pi(A) = 3 + \gamma \cdot (-2) = 3 - 2\gamma$$
$$(A, x, 1) \to (C) \implies V^\pi(A) \leftarrow R + \gamma V^\pi(C) = 1 + \gamma \cdot 0 = 1$$
$$(C) \implies V^\pi(C) \leftarrow 0 \text{ (absorbing zero-reward state)}$$

3. No, because, for instance, in state A both actions x and y are played at least once.

Exercise 8 (4 marks)

Consider a binary classifier, defined in terms of the threshold ξ:

$$y(\mathbf{x}) = \begin{cases} 1 & \text{if } \mathbf{w}^T\mathbf{x} \geq \xi \\ 0 & \text{otherwise} \end{cases},$$

where the weights $\mathbf{w}$ were trained on a dataset made of $N = 100$ samples and $\xi = 0$.

1. Suppose that Accuracy = Precision = 0.5 and that the negative samples correctly classified are 10, draw the confusion matrix.
2. Compute the Recall.
3. Suppose that we move the threshold ξ to the value 0.2. How will the Recall change?. Provide a formal justification to your answer (**Suggestion.** Function $\frac{x}{x+y}$ for $x, y \geq 0$ is increasing in x and decreasing in y)

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1. Knowing that Accuracy = Precision = 0.5, we deduce:

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} = \frac{TP+TN}{100} = 0.5 \implies TP+TN = 50,$$
$$\text{Precision} = \frac{TP}{TP+FP} = 0.5 \implies (1-0.5)TP = 0.5FP \implies TP = FP.$$

Thus, recalling that $TN = 10$, we have the system:

$$\begin{cases} TP+TN = 50 \\ TP = FP \\ TN = 10 \end{cases} \implies \begin{cases} TP = 40 \\ FP = 40 \\ TN = 10 \end{cases}.$$

From which, $FN = N - TP - FP - TN = 10$. Thus, the confusion matrix becomes:

	Actual Class: 1	Actual Class: 0
Predicted Class: 1	40	40
Predicted Class: 0	10	10

2. Using the confusion matrix, we ca compute the Recall:

$$\text{Recall} = \frac{TP}{TP+FN} = \frac{40}{40+10} = 0.8.$$

3. If we increase ξ (it does not matter the actual value 0.2), we will have that some points previously classified as positive, will be classified as negative. Which means, formally: $TP' \leq TP$, $FP' \leq FP$, $FN' \geq FN$ and $TN' \geq TN$, having denoted with $'$ the quantities after the change of ξ. The recall will become:

$$\text{Recall}' = \frac{TP'}{TP'+FN'} \leq \frac{TP}{TP+FN} = \text{Recall},$$

having applied the suggestion with $x = TP'$ and $y = FN'$ and observing that $TP' \leq TP$ and the function is increasing in x and $FN' \geq FN$ and the function is decreasing in y.